LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of basic block diagram of I-band transponder
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of basic block diagram of I-band transponder.
5. Specific Objectives	18.1.4.1 Trainee will be able to appraise the function of secondary radars in ASW operation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of transponder block diagram components; Safety: No live equipment handling; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a secondary radar system?
(b) Define the term 'I-band' in the context of radar frequencies.
(c) How does a secondary radar differ from a primary radar?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Pulse Repetition Frequency (PRF) • Pulse Width and Duty Cycle • Antenna Beam Pattern • Receiver Sensitivity • Transmitter Power Output	L & Ds	PPT, IB	15 min
(b)	• Intermediate Frequency (IF) Amplifier • Video Detector Function • Video Amplifier Gain • Video Limiter Circuit • Video Detector Output	L & Ds	PPT, PDF	20 min
(c)	• Video Limiter Output • Video Detector Output • Intermediate Frequency (IF) Amplifier • Video Amplifier Gain • Video Limiter Circuit	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the video limiter in the transponder block diagram.
(b) Describe how the I-band transponder contributes to Anti-Submarine Warfare (ASW) operations.

12. Home Assignment and Follow Up	Prepare a brief note on the application of Radar in surveillance operation for the next lesson (LP 09).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Brief description on application of Radar in surveillance operation. Features generally available on surveillance radar. Information obtained by surveillance radar. Concept of controlling and supporting aircraft taking part in surveillance operation and methods of passing information.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min